

Insert new annexes Annex 120F and Annex 120G corresponding to Clause 120 as follows:

Annex 120F

(normative)

Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.1 Overview

This annex defines the functional and electrical characteristics for the optional chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C). The 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces provide electrical characteristics and associated compliance points, which can optionally be used when designing systems with electrical interconnect of approximately 25 cm in length.

100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C are physical instantiations of the PMA service interface between the FEC (for 100GBASE-P PHYs) or PCS (for 200GBASE-R and 400GBASE-R PHYs) and the PMD, as described in 135.1.4 and 120.1.4. Figure 120F–1 shows example relationships of the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces to the ISO/IEC Open System Interconnection (OSI) reference model.

The sublayers including PCS and FEC (for 100GBASE-P PHYs) for each PHY that can optionally include a 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C are summarized in the tables in 80.1.5 and 116.1.4 and are specified in the corresponding PMD clause. The positioning of the 100GAUI-1 C2C, 200GAUI-2 C2C, or 400GAUI-4 C2C relative to other sublayers is shown in 135.1 and 120.1 with further examples in Annex 135A and Annex 120A.

The 100GAUI-1, 200GAUI-2, or 400GAUI-4 C2C bidirectional link is described in terms of a C2C transmitter, a C2C channel, and a C2C receiver. Figure 120F–2 depicts a typical C2C application. The C2C components at both ends of the link have connected ground references.

The C2C interface is composed of independent data paths in each direction. Each 100GAUI-1, 200GAUI-2, or 400GAUI-4 data path contains one, two, or four, respectively, differential lanes, which are AC-coupled.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C use PAM4 signaling. The nominal signaling rate for each lane is 53.125 GBd.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C channel is specified in 120F.4.

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter on each end of the link is adjusted to an appropriate setting, which may be based on channel knowledge. If implemented, the transmitter equalization feedback mechanism described in 120F.3.2.6 may be used to identify an appropriate setting of the link partner transmitter. The adaptive or adjustable receiver performs the remainder of the equalization.

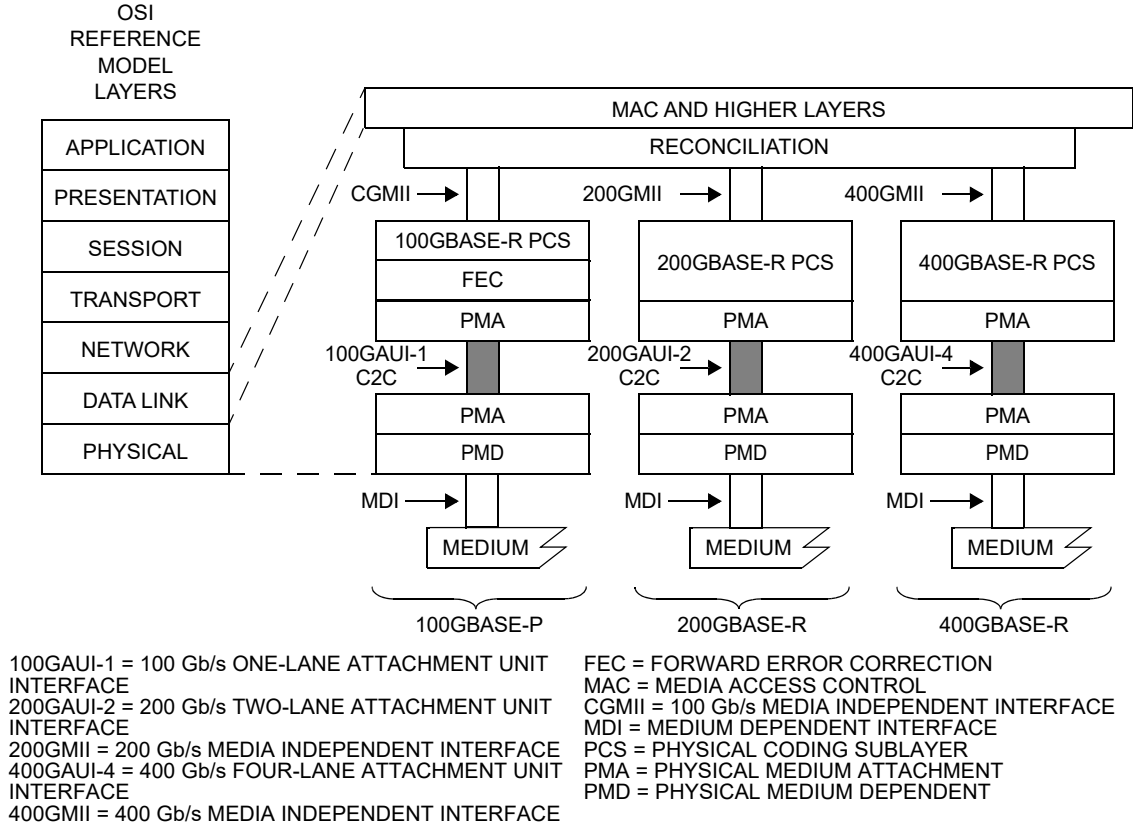
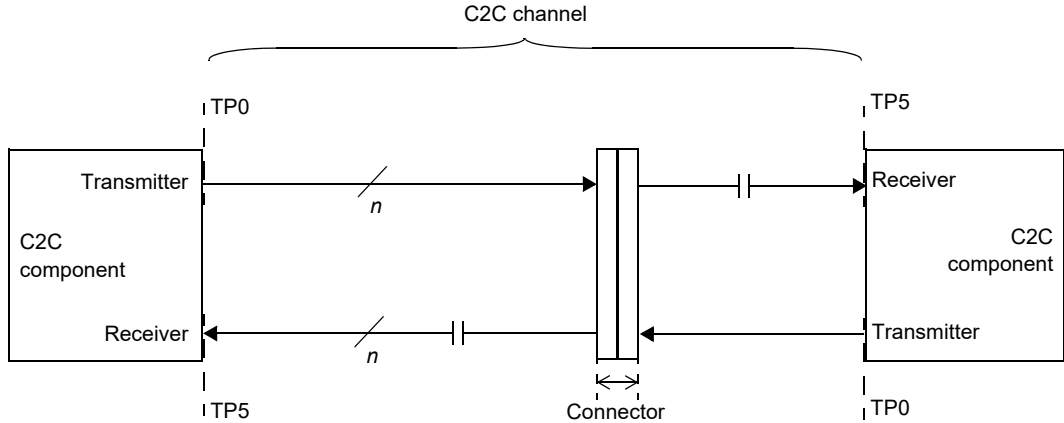


Figure 120F-1—Example 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C relationship to the ISO/IEC Open System Interconnection (OSI) reference model and the IEEE 802.3 Ethernet model



NOTE—The number of lanes n is equal to 1 for 100GAUI-1, 2 for 200GAUI-2, and 4 for 400GAUI-4.

Figure 120F-2—Typical 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C application

The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C transmitter supports $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2, that may be enabled or disabled as required. The 100GAUI-1, 200GAUI-2, and 400GAUI-4 C2C receiver may support $1/(1+D)$ mod 4 precoding, as specified in 135.5.7.2 and 120.5.7.2. Precoding may be enabled and disabled using the precoder request mechanism specified in 135F.3.2.1.

120F.2 Compliance points

The electrical characteristics for the 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces are defined at test points as described in 163.9.2.1 and 163.9.3.2.

120F.2.1 Reference impedance

The reference impedance for differential specifications is 100 Ω . The reference impedance for common-mode specifications is 25 Ω .

120F.3 Electrical characteristics

120F.3.1 Transmitter electrical characteristics

A 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C transmitter shall meet the specifications given in Table 120F–1.

Table 120F–1—Transmitter electrical characteristics at TP0v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)		53.125 ± 50 ppm ^a	GBd
Differential peak-to-peak output voltage ^b (max) Transmitter disabled Transmitter enabled	93.8.1.3	35 1200	mV mV
Common-mode voltage ^b (max)	93.8.1.3	1	V
Common-mode voltage ^b (min)	93.8.1.3	0.2	V
Low-frequency peak-to-peak AC common-mode voltage, $V_{CM_{LF}}$ (max)	120F.3.1.1	32	mV
Signal to AC common-mode noise ratio, $SCMR$ (min)	120F.3.1.2	15	dB
Difference effective return loss, $dERL$ (min)	120F.3.1.4	–3	dB
Common-mode to common-mode return loss, RL_{cc} (min)	162.9.4.9	2	dB
Difference steady-state voltage, dv_f (min)	163.9.2.4	0	V
Difference linear fit pulse peak ratio, dR_{peak} (min)	163A.3.2.1	0	—
Level separation mismatch ratio, R_{LM} (min)	162.9.4.2	0.95	—

Table 120F–1—Transmitter electrical characteristics at TP0v (*continued*)

Parameter	Reference	Value	Units
Output waveform ^c			
absolute value of step size for all taps (min)	162.9.4.1.4	0.005	—
absolute value of step size for all taps (max)	162.9.4.1.4	0.025	—
value at min state for c(–3) (max)	162.9.4.1.5	–0.05	—
value at max state for c(–3) (min)	162.9.4.1.5	0	—
value at min state for c(–2) (max)	162.9.4.1.5	0	—
value at max state for c(–2) (min)	162.9.4.1.5	0.1	—
value at min state for c(–1) (max)	162.9.4.1.5	–0.3	—
value at max state for c(–1) (min)	162.9.4.1.5	0	—
value at min state for c(0) (max)	162.9.4.1.5	0.5	—
value at min state for c(1) (max)	162.9.4.1.5	–0.1	—
value at max state for c(1) (min)	162.9.4.1.5	0	—
Signal-to-noise-and-distortion ratio, SNDR (min)	162.9.4.6	32.5	dB
Signal-to-residual-intersymbol-interference ratio, SNR_{ISI} (min)	120F.3.1.3	28	dB
Output jitter			
J_{RMS} (max)	120F.3.1.6	0.023	UI
J_{4u} (max)	120F.3.1.6	0.128	UI
$J_{4u_{03}}$ (max)	120F.3.1.6	0.118	UI
Even-odd jitter (max)	120F.3.1.6	0.025	UI

^a For a PMA in the same package as the PCS sublayer. In other cases, the signaling rate is derived from the signaling rate presented to the input lanes (see Figure 135–3 and Figure 120–3) by the adjacent PMD, PMA, or FEC sublayers.

^b Measurement uses the method described in 93.8.1.3 with the exception that the PRBS13Q test pattern is used.

^c The state of the transmit equalizer is controlled by management interface.

A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for all transmitter signal measurements, unless otherwise specified.

120F.3.1.1 Peak-to-peak AC common-mode voltage

The low-frequency and full-band peak-to-peak AC common-mode voltage, VC_{MF} and VC_{MB} , respectively, are defined by the method specified in 162.9.4.4 with the exception that the peak-to-peak AC common-mode voltage is defined as the AC common-mode voltage range measured at TP0v that includes all but 10^{-5} of the measured distribution, from 0.000005 to 0.999995 of the cumulative distribution.

The low-frequency peak-to-peak AC common-mode voltages shall meet the specification for VC_{MF} (max) in Table 120F–1.

120F.3.1.2 Signal to AC common-mode noise ratio

Signal to AC common-mode noise ratio, $SCMR$, is defined by the method specified in 163.9.2.6 with the exception that the full-band peak-to-peak AC common-mode voltage is defined in 120F.3.1.1.

The signal to AC common-mode noise ratio shall meet the specification for $SCMR$ (min) in Table 120F–1.

120F.3.1.3 Signal-to-residual-intersymbol-interference ratio

Signal-to-residual-intersymbol-interference ratio SNR_{ISI} is defined by the method specified in 162.9.4.3 with the exception that the continuous time filter settings are provided in Table 120F–8.

Signal-to-residual-intersymbol-interference ratio shall meet the specification for SNR_{ISI} (min) in Table 120F–1.

120F.3.1.4 Transmitter difference effective return loss (ERL)

The transmitter difference ERL at TP0v is computed using the procedure in 163A.3.2.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP0 to TP0v.

Table 120F–2—Transmitter and receiver ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	200	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Tukey window flag	tw	1	—

The difference ERL at TP0v shall be greater than or equal to $dERL$ (min) specified in Table 120F–1.

120F.3.1.5 Transmitter equalization settings

Each transmitter includes programmable equalization to compensate for the frequency-dependent loss of the channel and to facilitate data recovery at the receiver. The functional model for the transmit equalizer is the five-tap transversal filter shown in Figure 120F–3. Transmitter output waveform specifications are provided in Table 120F–1. The state of the transmitter equalizer may be configured via the transmitter control interface described in 120F.3.1.7.

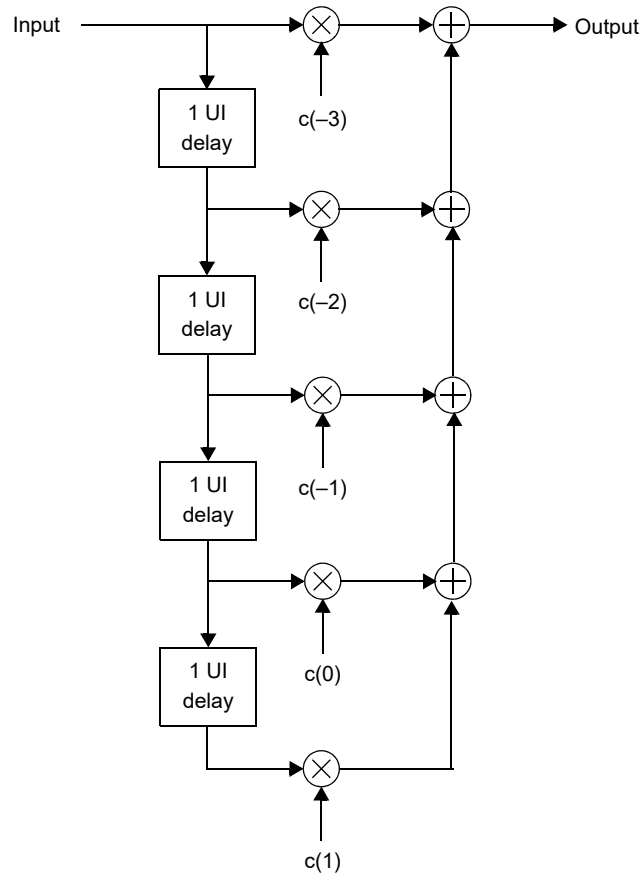


Figure 120F-3—Transmitter equalizer functional model

120F.3.1.6 Output jitter

Output jitter is characterized by four parameters: J_{4u} , $J_{4u_{03}}$, J_{RMS} , and even-odd jitter. These parameters are calculated from measurements with a single transmit equalizer setting to compensate for the loss of the transmitter package and test fixture. The equalizer setting is chosen to minimize any or all of the jitter parameters.

J_{4u} and J_{RMS} are calculated using the jitter measurement method specified in 120D.3.1.8.1. $J_{4u_{03}}$ is calculated the same way as J_{4u} except that the calculation uses only transitions R03 and F30 in Table 120D-4.

Even-odd jitter is calculated using the jitter measurement method specified in 162.9.4.7.

NOTE— J_{4u} is sensitive to measurement noise being converted to timing errors. Hence, accounting for measurement noise effects is recommended.

120F.3.1.7 Transmitter control

The transmitter output equalization is characterized using the linear fit method described in 162.9.4.1.1.

Transmitter equalization shall be configurable by management using a set of control and status variables listed in Table 120F–3, or an equivalent interface. The variables that control transmitter equalization are specific for each lane.

Table 120F–3—Transmitter control variables

Variable name	Description	Management access ^a
local_ic_req	Requested initial condition in the local transmitter.	RW
local_coef_sel	Coefficient selection in the local transmitter.	RW
local_coef_req	Update to the selected coefficient in the local transmitter.	RW
local_ic_sts	Status of initial condition request.	RO
local_k	Coefficient select echo.	RO
local_coef_sts	Coefficient update acknowledge.	RO

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–3 are the same as the corresponding ones without the “local_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter control is identical to the “LD” variable mapping in Table 162–7.

120F.3.2 Receiver characteristics

A receiver shall meet the specifications given in Table 120F–4, with precoding enabled on the test transmitter if required by the receiver.

Table 120F–4—Receiver characteristics at TP5v

Parameter	Reference	Value	Units
Signaling rate, each lane (range)	120F.3.2.1	53.125 ± 100 ppm	GBd
Difference effective return loss, <i>dERL</i> (min)	120F.3.2.2	–3	dB
Differential-mode to common-mode return loss, <i>RLcd</i>	120F.3.2.3	Equation (120F–1)	dB
Interference tolerance	120F.3.2.4	Table 120F–5	—
Jitter tolerance	120F.3.2.5	Table 162–17	—

120F.3.2.1 Receiver signaling rate

The receiver shall comply with the requirements of 120F.3.2.4 and 120F.3.2.5 for any signaling rate in the range $53.125 \text{ GBd} \pm 100 \text{ ppm}$.

120F.3.2.2 Receiver difference ERL

The receiver difference ERL at TP5v is computed using the procedure in 163A.4.1.2 with the values in Table 120F–2 and Table 120F–8, and with the value of T_{fx} equal to twice the delay from TP5 to TP5v.

Receiver difference ERL at TP5v shall be greater than or equal to $dERL$ (min) specified in Table 120F–4.

120F.3.2.3 Receiver differential-mode to common-mode return loss

The differential-mode to common-mode return loss of the receiver shall meet Equation (120F–1). The differential-mode to common-mode return loss limit is illustrated by Figure 120F–4.

$$RL_{cd}(f) \geq \begin{cases} 25 - 0.72f & 0.05 \leq f \leq 13.9 \\ 15 & 13.9 < f \leq 38 \end{cases} \quad (120F-1)$$

where

RL_{cd} is the differential-mode to common-mode return loss in dB
 f is the frequency in GHz

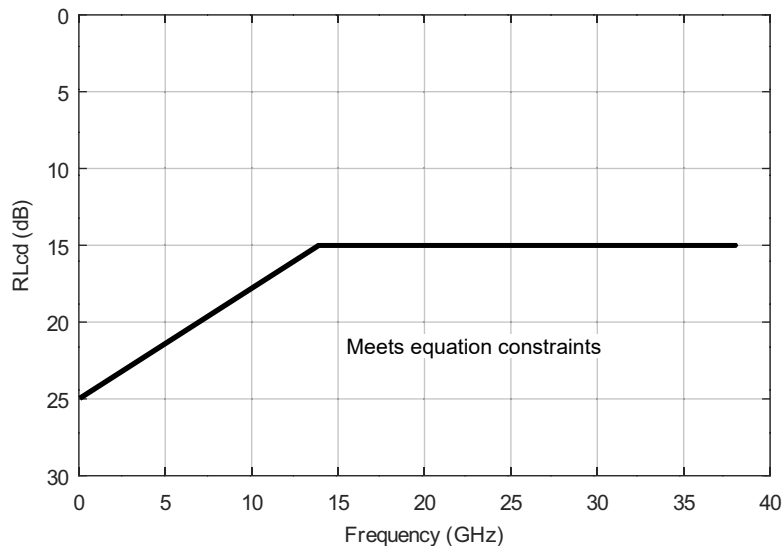


Figure 120F–4—Receiver differential-mode to common-mode return loss limit

120F.3.2.4 Receiver interference tolerance

Receiver interference tolerance is defined by the procedure in [Annex 93C](#) with the exception that transmitter equalization is configured by management (see [120D.3.2.3](#)) to the settings that provide the lowest FEC symbol error ratio. The receiver on each lane shall meet the FEC symbol error ratio requirement with

channels matching the Channel Operating Margin (COM) and loss parameters for Test 1 and Test 2 in Table 120F–5. The following additional considerations apply to the interference tolerance test.

- a) The test transmitter is constrained such that for any transmitter equalizer setting the differential peak-to-peak voltage (see 93.8.1.3) is less than or equal to 800 mV.
- b) The ERL of the test channel measured at TP5 replica towards TPt meets the requirements in 120F.4.3.
- c) The lower frequency bound for the noise spectral density constraints, f_{NSDI} , is 1 GHz.
- d) For the calculation of COM, the transmitter model is determined according to step e in 163.9.3.5.
- e) For the calculation of test channel COM, the following parameters are based on values measured from the test transmitter.
 - The parameter SNR_{TX} is set to the measured value of SNDR with $N_p=11$.
 - The parameter R_{LM} is set to the measured value of R_{LM} .
 - The parameters A_{DD} and σ_{RJ} are calculated from the measured values of J_{4u} and J_{RMS} using Equation (120D–10), and Equation (120D–11), respectively, with the exceptions that $Q4$ is replaced with $Q4d$ and $Q4d$ is equal to 3.719.
- f) Other COM parameters are set according to the values in Table 120F–8.
- g) COM is calculated using both Test 1 and Test 2 device package model transmission line lengths listed in Table 120F–8 on the receiver side. The value of COM is taken as the lower of the two calculated values.
- h) The test pattern is the scrambled idle test pattern defined in 119.2.4.9.
- i) As an alternative to using the scrambled idle test pattern and measuring FEC symbol error ratio it is permissible to use the PRBS31Q pattern as described in 120.5.11.2.2 and bit error ratio testing. In this case the required bit error ratio is equal to the required FEC symbol error ratio divided by 10. Note that this requirement can be somewhat more stringent than using the scrambled idle test pattern and measuring FEC symbol error ratio, and therefore failing this test requirement with the PRBS31Q pattern does not necessarily imply a failure of the interference tolerance test.
- j) A test system with a fourth-order Bessel-Thomson low-pass response with 40 GHz 3 dB bandwidth is to be used for measurement of the signal applied by the pattern generator and for measurements of the broadband noise.

Table 120F–5—Receiver interference tolerance parameters

Parameter	Test 1 (low loss)			Test 2 (high loss)			Units
	Min	Max	Target	Min	Max	Target	
FEC symbol error ratio ^a	—	10^{-4}	—	—	10^{-4}	—	—
Insertion loss at 26.5625 GHz ^b	9.5	10.5	—	19.5	20.5	—	dB
RSS_DFE4 ^c	0.05	—	—	0.05	—	—	—
COM including effects of broadband noise ^d	—	—	3	—	—	3	dB

^a The FEC symbol error ratio is measured in step 10) of the receiver interference tolerance method defined in 93C.2

^b Measured between TPt and TP5 (see Figure 93C–4).

^c RSS_DFE4 is described in 93A.2.

^d The COM value is the target for the receiver noise level calibration defined in 93C.2 step 7). The channel noise voltage applied in 93C.2 step 8) should be as close as practical to the value needed to produce the target COM. If higher amplitude values are used, this would demonstrate margin to the specification but this is not required for compliance.

120F.3.2.5 Receiver jitter tolerance

Receiver jitter tolerance is verified for each pair of jitter frequency and peak-to-peak amplitude values listed in Table 162–17. The test setup shown in Figure 93–12, or its equivalent, is used. The test channel meets the insertion loss requirement for Test 2 in Table 120F–5. The synthesizer frequency is set to the specified jitter frequency and the synthesizer output amplitude is adjusted until the specified peak-to-peak jitter amplitude for that frequency is measured at TP0v. The test procedure is the same as the one described in 120F.3.2.4, with the following exceptions:

- No broadband noise is added.
- The test channel COM, calculated per items d) through g) in 120F.3.2.4, is at least 3 dB.
- For the COM parameter calibration described in 120F.3.2.4 item e), the test channel transmitter J_{RMS} and J_{4u} values are measured with the jitter frequency and amplitude set according to Case F from Table 162–17.

The receiver under test shall meet the FEC symbol error ratio in Table 120F–5 for each case in Table 162–17.

120F.3.2.6 Transmitter equalization feedback (optional)

Transmitter equalization feedback is an optional capability for a receiver. If implemented, it shall operate as described in this subclause.

A receiver may generate a request to change the transmit equalization coefficients of the remote transmitter using the variables in Table 120F–6, or an equivalent interface.

Table 120F–6—Transmitter equalization feedback variables

Variable name	Description	Management access ^a
remote_ic_req	Initial condition request for the remote transmitter.	RO
remote_coef_sel	Coefficient selection for the remote transmitter.	RO
remote_coef_req	Requested update to the selected coefficient for the remote transmitter.	RO
remote_ic_sts	Status of initial condition request from the remote transmitter.	RW
remote_k	Coefficient select echo from the remote transmitter.	RW
remote_coef_sts	Coefficient update acknowledge from the remote transmitter.	RW

^a RW = Read/Write, RO = Read only.

Definitions of the variables in Table 120F–6 are the same as the corresponding ones without the “remote_” prefix in 136.8.11.7.1, with the exceptions listed in 162.8.11, and the additional exception that all variables are accessed and controlled by management instead of being mapped to training frame fields.

The requested setting is generated from information internal to the receiver in an implementation specific manner.

If a Clause 45 MDIO is implemented, the variable mapping for the transmitter equalization feedback is identical to the “LP” variable mapping in Table 162–7.

120F.4 Channel characteristics

Table 120F–7 provides a summary of the channel characteristics for 100GAUI-1 C2C, 200GAUI-2 C2C, and 400GAUI-4 C2C interfaces, and references to the subclauses addressing each parameter.

Table 120F–7—Channel characteristics summary

Description	Reference	Value	Unit
Minimum COM	120F.4.1	3	dB
Maximum insertion loss at 26.56 GHz (recommended)	120F.4.2	20	dB
Minimum ERL	120F.4.3	9.7	dB
Differential-mode to common-mode return loss, RL_{cd}	120F.4.4	Equation (163–6)	dB
Maximum AC-coupling 3 dB corner frequency	120F.4.5	50	kHz

120F.4.1 Channel operating margin

The Channel Operating Margin (COM) is computed using the procedure in 93A.1, where T_r is 7.5 ps for $H_f(f)$ in Equation (93A–19), and the parameters in Table 120F–8. COM is calculated twice, Test 1 and Test 2, with different values of device package model transmission line length z_p . COM shall be greater than or equal to 3 dB for both Test 1 and Test 2. This minimum value allocates margin for practical limitations on the receiver implementation as well as the allowed transmitter equalization coefficients.

Table 120F–8—COM parameter values

Parameter	Symbol	Value	Units
Signaling rate	f_b	53.125	GBd
Maximum start frequency	$f_{\min}$	0.05	GHz
Maximum frequency step	Δf	0.01	GHz
Device package model			
Single-ended device capacitance	C_d	$1.2e-4$	nF
Single-ended device series inductance	L_s	0.12	nH
Single-ended bump capacitance	C_b	3×10^{-5}	nF
Transmission line length, Test 1 Tx	z_p	13	mm
Transmission line length, Test 1 Rx	z_p	11	mm
Transmission line length, Test 2 Tx	z_p	31	mm
Transmission line length, Test 2 Rx	z_p	29	mm
Transmission line 2 length	z_{p2}	1.8	mm
Transmission line parameter, a_1	a_1	9.909×10^{-4}	ns ^{1/2} /mm
Transmission line parameter, a_2	a_2	2.772×10^{-4}	ns/mm
Single-ended package capacitance at package-to-board interface	C_p	8.7×10^{-5}	nF
Package transmission line nominal characteristic impedance	Z_c	87.5	Ω
Package transmission line 2 nominal characteristic impedance	Z_{c2}	92.5	Ω
Single-ended reference resistance	R_0	50	Ω
Single-ended termination resistance	R_d	50	Ω
Receiver 3 dB bandwidth	f_r	$0.75 \times f_b$	GHz

Table 120F–8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Transmitter equalizer, minimum cursor coefficient	$c(0)$	0.54	—
Transmitter equalizer, 3 rd pre-cursor coefficient Minimum value Maximum value Step size	$c(-3)$	−0.04 0 0.02	—
Transmitter equalizer, 2 nd pre-cursor coefficient Minimum value Maximum value Step size	$c(-2)$	0 0.1 0.02	—
Transmitter equalizer, 1 st pre-cursor coefficient Minimum value Maximum value Step size	$c(-1)$	−0.28 0 0.02	—
Transmitter equalizer, post-cursor coefficient Minimum value Maximum value Step size	$c(1)$	−0.1 0 0.02	—
Continuous time filter, DC gain Minimum value Maximum value Step size	g_{DC}	−20 0 1	dB dB dB
Continuous time filter, DC gain 2 Minimum value Maximum value Step size	g_{DC2}	−4 0 1	dB dB dB
Continuous time filter, zero frequency for $g_{DC} = 0$	f_z	$f_b / 2.5$	GHz
Continuous time filter, pole frequencies	f_{p1} f_{p2}	$f_b / 2.5$ f_b	GHz GHz
Continuous time filter, low-frequency pole/zero	f_{LF}	$f_b / 80$	GHz
Transmitter differential peak output voltage Victim Far-end aggressor Near-end aggressor	A_v A_{fe} A_{ne}	0.413 0.413 0.608	V V V
Number of signal levels	L	4	—
Level separation mismatch ratio	R_{LM}	0.95	—
Transmitter signal-to-noise ratio	SNR_{TX}	33	dB
Number of samples per unit interval	M	32	—
Decision feedback equalizer (DFE) length	N_b	6	UI
Normalized DFE coefficient maximum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\max}(n)$	0.65 0.15 0.1	—
Normalized DFE coefficient minimum limit $n = 1$ $n = 2$ $n = 3$ to N_b	$bb_{\min}(n)$	0.3 0.05 −0.04	—

Table 120F-8—COM parameter values (*continued*)

Parameter	Symbol	Value	Units
Random jitter, RMS	σ_{RJ}	0.01	UI
Dual-Dirac jitter, peak	A_{DD}	0.02	UI
One-sided noise spectral density	η_0	2×10^{-8}	V ² /GHz
Target detector error ratio	DER_0	10^{-5}	—

120F.4.2 Channel insertion loss (recommended)

The channel insertion loss should meet Equation (120F-2). Actual channel insertion loss could be higher or lower than that given by Equation (120F-2) due to the channel insertion loss deviation (ILD), return loss, and crosstalk. Note that for this equation the channel insertion loss at the Nyquist frequency is less than or equal to 20 dB.

$$ILdd(f) \leq 1.083 + 1.444\sqrt{f} + 0.432f \quad (120F-2)$$

for $0.01 \leq f \leq 53.125$

where

f is the frequency in GHz
 $ILdd$ is the channel insertion loss in dB

The insertion loss limit is illustrated by Figure 120F-5.

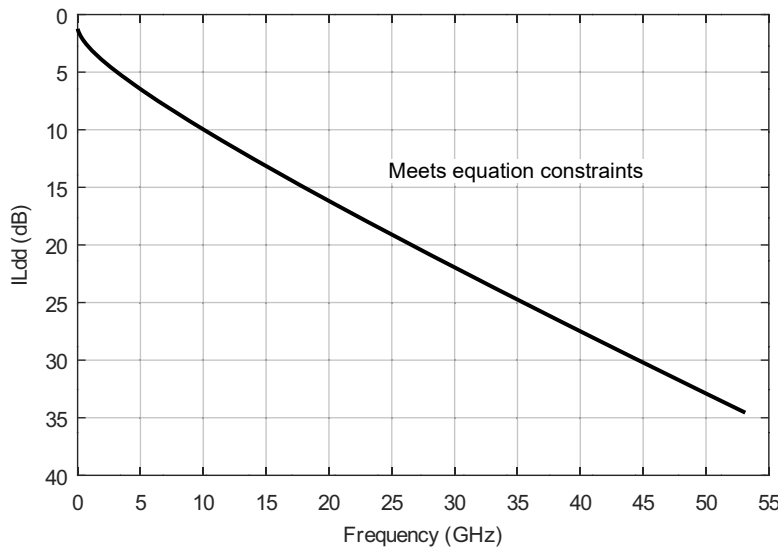


Figure 120F-5—Channel insertion loss limit

120F.4.3 Channel ERL

The channel ERL at TP0 and at TP5 is computed using the procedure in 93A.5 with the values in Table 120F–9. Parameters that do not appear in Table 120F–9 take values from Table 120F–8. Channel ERL at TP0 and at TP5 shall be greater than or equal to 9.7 dB.

Table 120F–9—Channel ERL parameter values

Parameter	Symbol	Value	Units
Transition time associated with a pulse	T_r	0.01	ns
Incremental available signal loss factor	β_x	0	GHz
Permitted reflection from a transmission line external to the device under test	ρ_x	0.618	—
Length of the reflection signal	N	2000	UI
Equalizer length associated with reflection signal	N_{bx}	6	UI
Time-gated propagation delay	T_{fx}	0	ns
Tukey window flag	tw	1	—

120F.4.4 Channel differential-mode to common-mode return loss

The TP0 to TP5 channel differential-mode to common-mode return loss shall meet Equation (163–6) illustrated in Figure 163–7.

120F.4.5 Channel AC-Coupling

Channel AC-coupling is specified in 163.10.7.

120F.5 Protocol implementation conformance statement (PICS) proforma for Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)²⁰

120F.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in [Clause 21](#).

120F.5.2 Identification

120F.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120F.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3ck-2022, Annex 120F, Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21 ; the answer Yes means that the implementation does not conform to IEEE Std 802.3ck-2022.)	

Date of Statement	
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²⁰Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120F.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
NOL	Number of differential AC-coupled lanes	120F.1	One independent data path in each direction for 100GAUI-1 C2C, two independent data paths in each direction for 200GAUI-2 C2C, and four independent data paths in each direction for 400GAUI-4 C2C	M	Yes []
*CHAN	Channel	120F.4	Items marked with CHAN include channel specifications not applicable to a PHY manufacturer	O	Yes [] No []
*PCSP KG	PMA in same package as PCS.			O	Yes [] No []

120F.5.4 PICS proforma tables for Chip-to-chip 100 Gb/s one-lane Attachment Unit Interface (100GAUI-1 C2C), 200 Gb/s two-lane Attachment Unit Interface (200GAUI-2 C2C), and 400 Gb/s four-lane Attachment Unit Interface (400GAUI-4 C2C)

120F.5.4.1 Transmitter

Item	Feature	Subclause	Value/Comment	Status	Support
TC1	Signaling rate, each lane, PMA is in same package as PCS	120F.3.1	Per Table 120F–1	PCSP KG:M	Yes [] N/A []
TC2	Signaling rate, each lane, PMA is not in same package as PCS	120F.3.1	Derived from PMA input lanes	!PCSP KG:M	Yes [] N/A []
TC3	Peak-to-peak differential output voltage, transmitter enabled	120F.3.1	Per Table 120F–1	M	Yes []
TC4	Peak-to-peak differential output voltage, transmitter disabled	120F.3.1	Per Table 120F–1	M	Yes []
TC5	DC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC6	Low-frequency peak-to-peak AC common-mode voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC7	Signal to AC common-mode noise ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC8	Difference ERL	120F.3.1	Per Table 120F–1	M	Yes []
TC9	Common-mode to common-mode return loss	120F.3.1	Per Table 120F–1	M	Yes []
TC10	Output waveform coefficient limits	120F.3.1	Per Table 120F–1	M	Yes []

Item	Feature	Subclause	Value/Comment	Status	Support
TC11	Output waveform coefficient step size	120F.3.1	Per Table 120F–1	M	Yes []
TC12	Output jitter	120F.3.1	Per Table 120F–1	M	Yes []
TC13	Signal-to-noise-and-distortion ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC14	Signal-to-residual-intersymbol-interference ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC15	Transmitter precoder request	120F.1		O	Yes [] No []
TC16	Transmit equalizer controlled by management interface	120F.3.1.7		M	Yes []
TC17	Difference steady-state voltage	120F.3.1	Per Table 120F–1	M	Yes []
TC18	Difference linear fit pulse peak ratio	120F.3.1	Per Table 120F–1	M	Yes []
TC19	Level separation mismatch ratio	120F.3.1	Per Table 120F–1	M	Yes []

120F.5.4.2 Receiver

Item	Feature	Subclause	Value/Comment	Status	Support
RC1	Difference ERL	120F.3.2	Per Table 120F–4	M	Yes []
RC2	Differential-mode to common-mode return loss	120F.3.2	Per Table 120F–4	M	Yes []
RC3	Interference tolerance	120F.3.2.4		M	Yes []
RC4	Jitter tolerance	120F.3.2.5		M	Yes []
RC5	Transmitter equalization feedback	120F.3.2.6		O	Yes [] No []

Channel

Item	Feature	Subclause	Value/Comment	Status	Support
CC1	Channel operating margin (COM)	120F.4.1	≥ 3 dB	CHAN :M	Yes [] N/A []
CC2	Channel insertion loss	120F.4.2	Meets Equation (120F–2) constraints	CHAN :O	Yes [] No [] N/A []
CC2	Channel ERL	120F.4.3	≥ 9.7 dB	CHAN :M	Yes [] N/A []
CC4	Channel AC-coupling	120F.4.5	Per 163.10.7	CHAN :M	Yes [] N/A []